# Ayush Maheshwari

Senior Solutions Architect, NVIDIA | Ph.D. [CSE, IIT Bombay] | Foundation Models

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## Summary

Ph.D. in ML from CSE, IIT Bombay and Senior Solutions Architect at NVIDIA, working on foundation models, multilingual NLP, Indic LLMs, distributed training, and model evaluation. Previously led Indic LLM development from scratch at Vizzhy, including training on large accelerator cluster.

## Experience

**NVIDIA** **Gurugram, India**
*Senior Solutions Architect, NVAITC* *Oct 2024 – Present*

- ◇ Working on foundation-model research across multilingual, domain-specific, and scientific AI problems, including data design, model adaptation, evaluation, and deployment on computing platforms.
- ◇ Collaborating with academic researchers, research organizations, and external AI teams on large-scale model development and scientific AI workloads.
- ◇ Reviewing large-scale training and inference setups, and debugging performance issues across data, model, compute, and serving layers.

**Vizzhy Inc** **Bengaluru, India**
*Lead Research Scientist, Language Model Team* *Sep 2023 – Sep 2024*

- ◇ Led a 5-member research team that built Indic large language models from scratch, covering data curation, tokenizer training, architecture choices, pretraining, post-training, and evaluation.
- ◇ Trained Indic LLMs on a 1,024 accelerator cluster, handling distributed training, experiment runs, checkpointing, failure recovery, throughput debugging, and model-quality analysis.

## Education

**Indian Institute of Technology Bombay** **Mumbai, India**
*Ph.D. in ML* *Jan 2019 – Aug 2023*
*Defence: July 2024*; '**Impactful Research Award 2023**' by IIT Bombay

**Thesis:** Knowledge Integration in NLP Models using Constraint Ingestion and Generation
*Advisor:* Prof. Ganesh Ramakrishnan

**Tata Institute of Social Sciences** **Mumbai, India**
*M.Sc.* *2013 – 2015*

**University College of Engineering** **Rajasthan Technical University Kota, India**
*B.Tech. (CSE)* *2007 – 2011*

## Skills

**LLM Systems**: Foundation Models, Distributed Training, LLM Pretraining, Post-training, Instruction Tuning, Model Adaptation, Inference Optimization and Model Deployment

**NLP**: Multilingual NLP, Low-Resource Languages, Neural Machine Translation, Semi-Supervised Learning

**Programming Languages/Tools**: Python, PyTorch, Megatron, Bash, SQL, NumPy, scikit-learn, C, C++, JavaScript

## Open-Source contributions

- **SPEAR** : Data-Programming python library to create training data. GitHub
- **UDAAN** : Post-editing workbench to quickly create publishable translations. Website

## Additional Experience

**Adobe Research** — **Bengaluru, India**
*Research Intern, Document Cloud Team* — *May 2021 – Aug 2021*

**NCETIS, IIT Bombay** — **Mumbai, India**
*Project Engineer* — *May 2016 – Dec 2018*

**Water & Climate Lab, IIT Gandhinagar** — **Gandhinagar, India**
*Junior Research Fellow* — *May 2015 – Apr 2016*

**Tata Consultancy Services** — **Mumbai, India**
*Systems Engineer* — *Oct 2011 – Jun 2013*

## Research Interests

- Develop large language models for mid and low-resource languages.
- Develop machine learning based methods that generalize to new domains with limited resources.
- Lexically constrained neural machine translation, data programming, semi-supervised learning

## Awards, Services & Talks

- My PhD thesis work was awarded **Impactful Research Award 2023** by IIT Bombay
- Best Demo Paper Award at CODS-COMAD 2023
- Awarded prestigious Ekal Fellowship 2020
- Reviewer: ARR 22–24, EACL, EMNLP, ACL (21–23)
- Fully covered grant for attending EMNLP 2022, Abu Dhabi
- Co-Founder at non-profit Temples of India - A web and app platform listing $> 700$K temples across Indian subcontinent.
- Key Member of Udaan Project - A project to translate education textbooks in Indian languages.

## Computer Skills

**Programming Languages**: Python, C, C++, Javascript/jQuery, Bash, SQL

**Libraries**: PyTorch, DeepSpeed, HuggingFace, NumPy, scikit-learn

## Talks

- Delivered Masterclass on **Building data-efficient large language models**
  NLTF NASSCOM, 2024
- Invited Tutorial on **Data Efficient Machine Learning for end-to-end Educational Content Creation**
  International Conference on Educational Data Mining, 2023
- Talk on **SPEAR: Data Programming using Semi-Supervision** [Talk]
  DECILE Workshop, 2022

## Publications

1. AQBENCH: Benchmarking Neural Surrogates for Air Quality Forecasting

*NeurIPS (ED Track), 2026*
Siddharthan D, Bedi S, Parmar P, **Maheshwari A**, Kota S, Krishnan N

2. IndicParam: Benchmark to evaluate LLMs on low-resource Indic languages
*EMNLP (Main), 2026*
**Maheshwari, A**, Sharma K, Patel V, Maheshwari A

3. LexGen: Domain-aware Multilingual Lexicon Generation
*ACL (Main) 2025*
**Maheshwari, A**, Singh A, Karthika NJ, Bhat K, Jyothi P, Ramakrishnan G

4. ARISE: Iterative Rule Induction and Synthetic Data Generation for Text Classification
*Findings of NAACL, 2025*
Yashwanth M, Singh V, **Maheshwari, A**, Krishna A, Ramakrishnan G

5. DictDis: Dictionary Constrained Disambiguation for Improved NMT
*Findings of EMNLP, 2024*
**Maheshwari, A**, Jyothi P, Ramakrishnan G

6. FAIR: Filtering of Automatically Induced Rules
*EACL 2024*
Bajpai, D.J, **Maheshwari, A**, Hanawal, M. and Ramakrishnan, G

7. Samayik: A Benchmark and Dataset for English-Sanskrit Translation
*LREC-COLING 2024*
**Maheshwari, A**, Gupta, A, Krishna, A, Singh, A.K, Ramakrishnan, G, Gourishetty, A.K. and Singla, J

8. Adaptive mixing of auxiliary losses in supervised learning
*AAAI, 2023* (Oral Presentation)
Sivasubramanian D, **Maheshwari, A**, Prathosh AP, Shenoy P, Ramakrishnan G

9. UDAAN-Machine Learning based Post-Editing tool for Document Translation
*CODS-COMAD Demonstrations, 2023* (Best Demo Paper Award)
**Maheshwari, A**, R. Ajay, Subramanian V, Ramakrishnan G

10. Eigen: Expert-Informed Joint Learning Aggregation for High-Fidelity Information Extraction from Document Images
*Machine Learning for Health (ML4H) at NeurIPS 2023*
Singh, A, Venkatapathy, **Maheshwari, A**, Narayan, P, Shetty, D.P. and Ramakrishnan, G

11. Learning to Robustly Aggregate Labeling Functions for Semi-supervised Data Programming
*Findings of ACL, 2022*
**Maheshwari, A**, Killamsetty K, Ramakrishnan G, Iyer R, Danilevsky M, Popa L

12. A Benchmark and Dataset for Post-OCR text correction in Sanskrit
*Findings of EMNLP, 2022*
**Maheshwari, A**, Singh, N, Krishna, A. and Ramakrishnan G,

13. Data Programming using Semi-Supervision and Subset Selection
*Findings of ACL, 2021*
**Maheshwari, A,** Chatterjee O, Killamsetty K, Iyer R, Ramakrishnan G

14. Joint Learning of Hyperbolic Label Embeddings for Hierarchical Multi-label Classification
*EACL 2021*
**Maheshwari A**, Chatterjee S, Ramakrishnan, G. Jagaralpudi S.

15. OCR On-the-Go: Robust End-to-end Systems for Reading License Plates & Street Signs
*International Conference on Document Analysis and Recognition, 2019*
Saluja, R, **Maheshwari, A**, Ramakrishnan, G, Chaudhuri, P. and Carman, M

16. DynGAN: Generative Adversarial Networks for Dynamic Network Embedding
*Graph Representation Learning Workshop at NeurIPS 2019*
**Maheshwari, A**, Goyal, A, Hanawal, M.K. and Ramakrishnan, G

17. Semi-supervised data programming in python
*EMNLP 2022 - System Demonstrations*
Abhishek, G, Ingole, H, Laturia, P, Dorna, V, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R,

18. Rule Augmented Unsupervised Constituency Parsing
*Findings of ACL, 2021*
Sahay, A, Nasery, A, **Maheshwari, A**, Ramakrishnan, G. and Iyer, R

19. Representation learning on graphs by integrating content and structure information
*11th International Conference on Communication Systems & Networks (COMSNETS)*
**Maheshwari, A**, Goyal, A, Kumar, A, Hanawal, M.K. and Ramakrishnan, G

20. Enhancing Low-Resource NMT with a Multilingual Encoder and Knowledge Distillation: A Case Study
*LoResMT Workshop at ACL 2024*
Roy, A, Ray, P, **Maheshwari, A**, Sarkar, S. and Goyal, P

21. Unsupervised learning of explainable parse trees for improved generalisation
*IJCNN 2021*
Sahay, A, **Maheshwari, A**, Kumar, R, Ramakrishnan, G, Hanawal, M.K. and Arya, K

22. Tale of tails using rule augmented sequence labeling for event extraction
*Star-AI Workshop at AAAI 2019*
**Maheshwari, A**, Patel, H, Rathod, N, Kumar, R, Ramakrishnan, G. and Bhattacharyya, P

23. Entity Resolution and Location Disambiguation in the Ancient Hindu Temples Domain using Web Data
*NAACL 2018 - System Demonstrations*
**Maheshwari, A**, Kumar, V, Ramakrishnan, G. and Nath, J.S